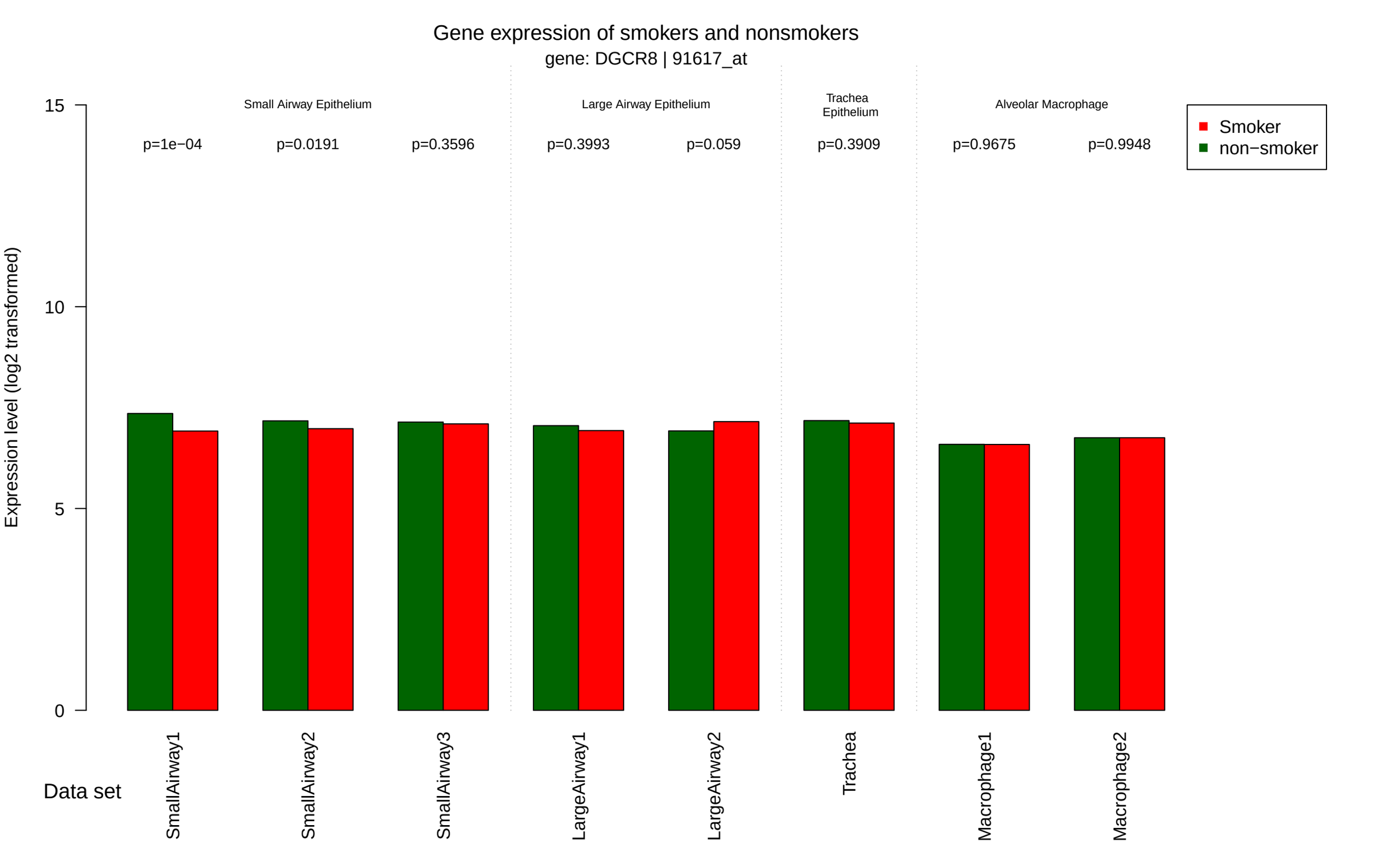
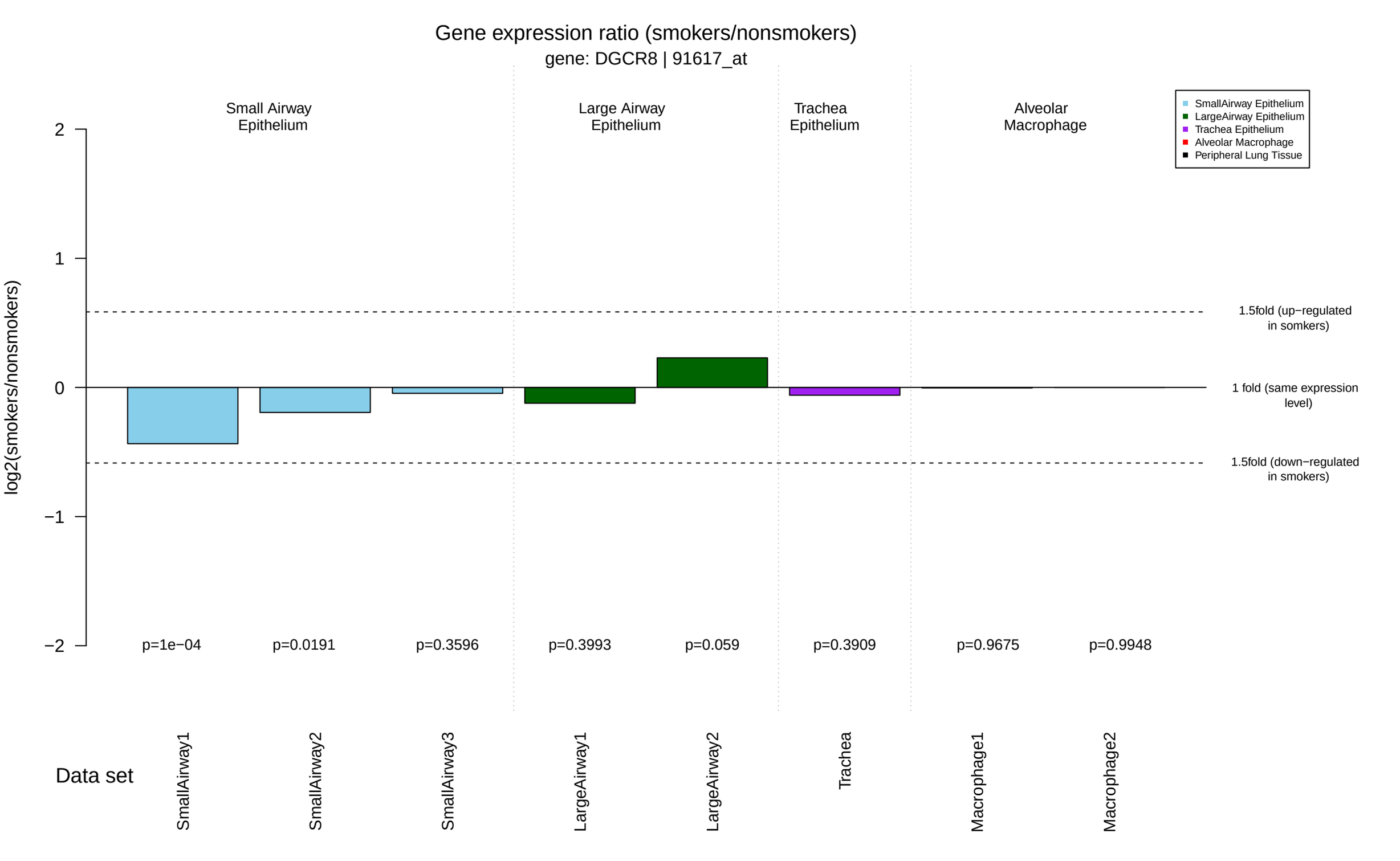
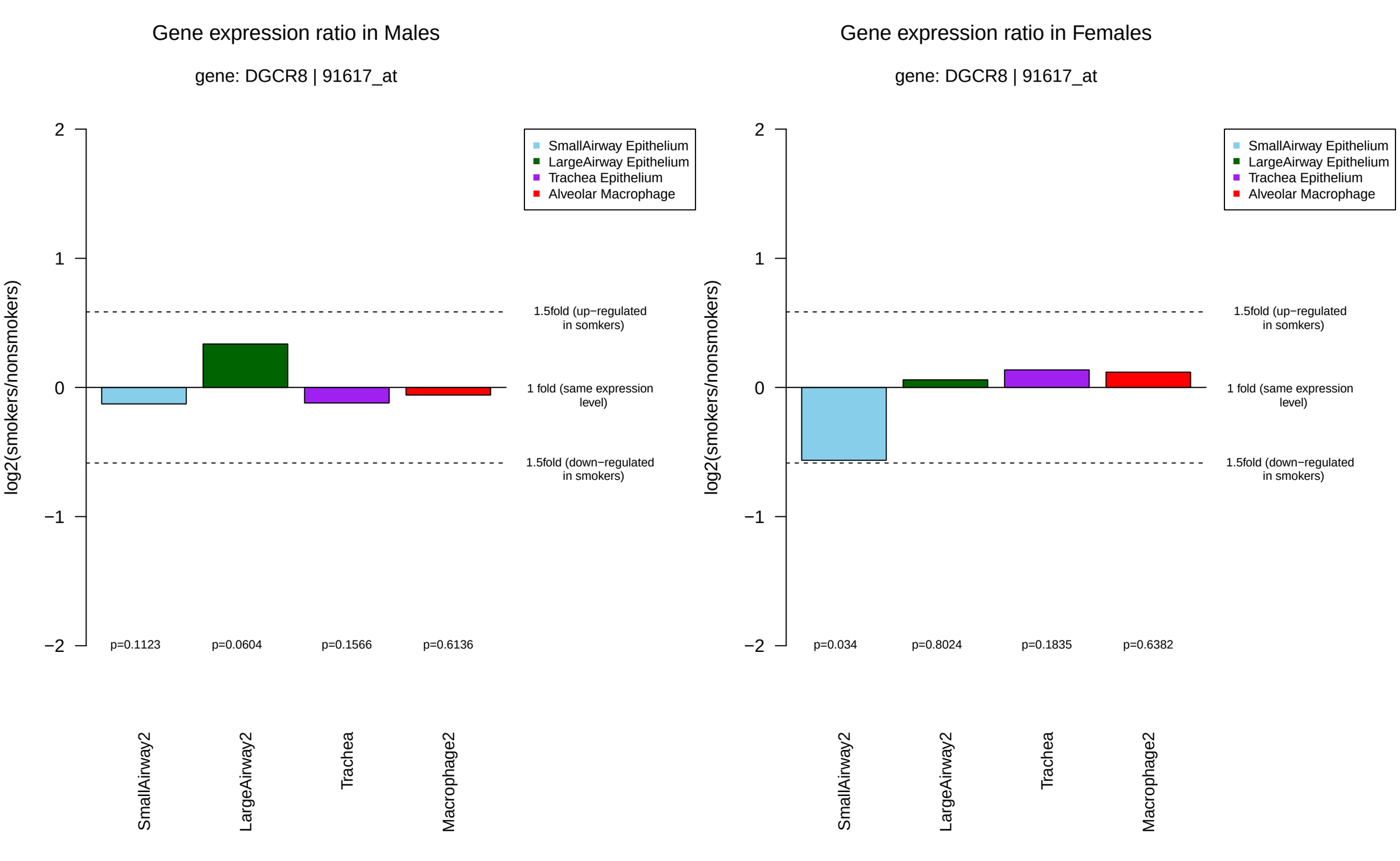
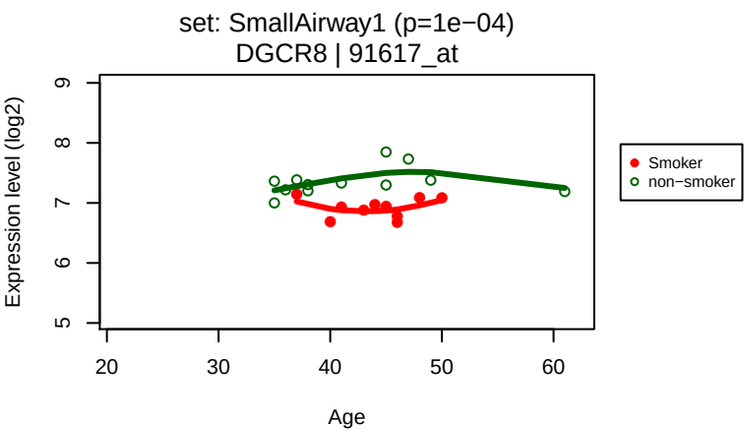
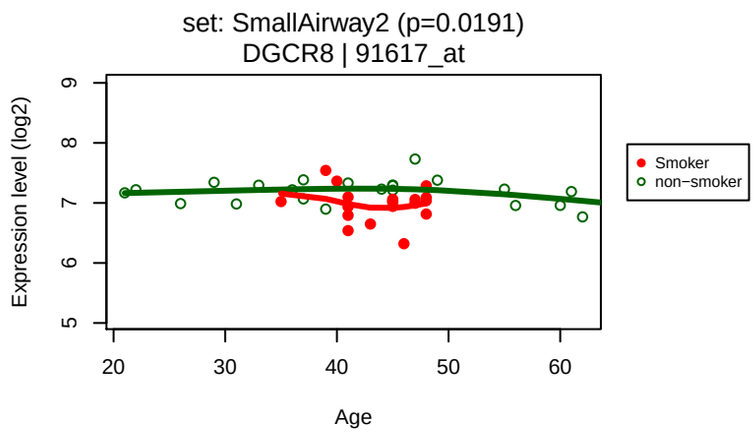
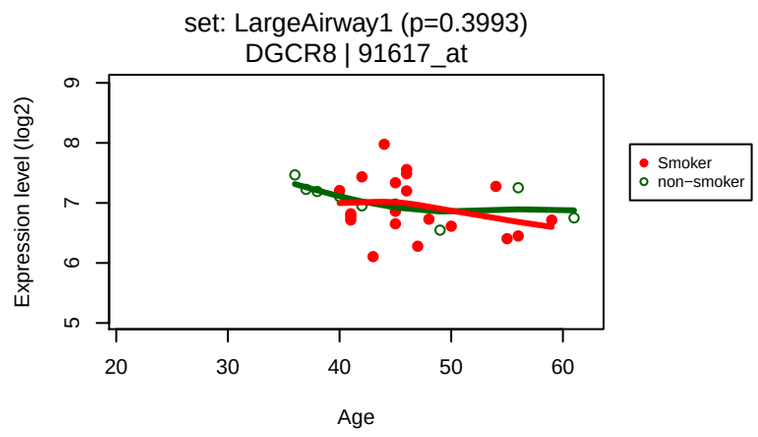

Gene expression of smokers and nonsmokers
gene: DGCR8 | 91617_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=1e−04
p=0.0191
p=0.3596
p=0.3993
p=0.059
p=0.3909
p=0.9675
p=0.9948
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

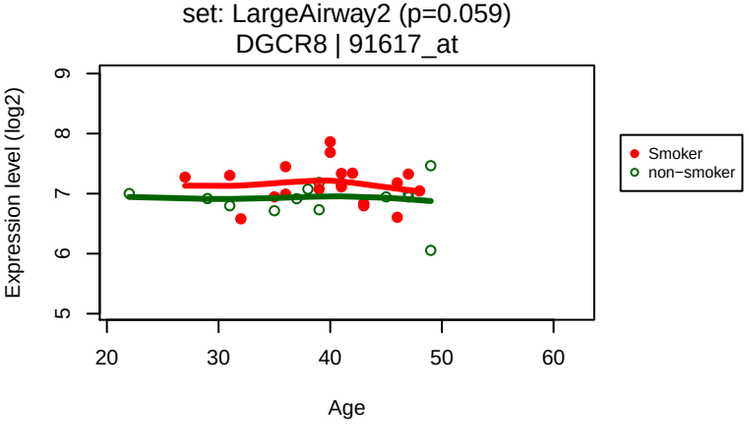

Gene expression ratio (smokers/nonsmokers)
gene: DGCR8 | 91617_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=1e−04
p=0.0191
p=0.3596
p=0.3993
p=0.059
p=0.3909
p=0.9675
p=0.9948
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

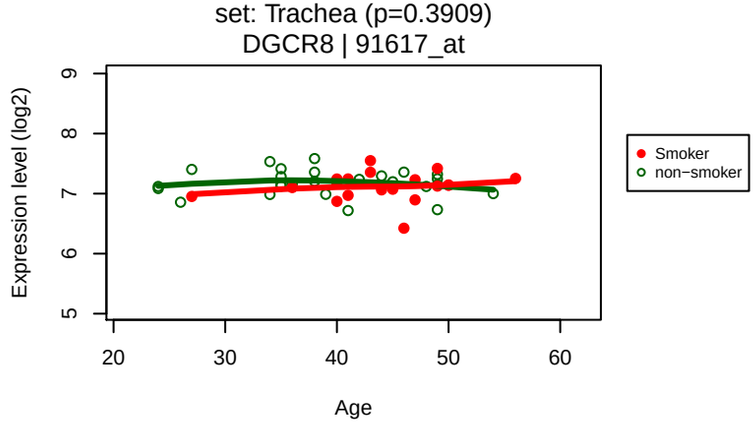

Gene expression ratio in Males
gene: DGCR8 | 91617_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1123
p=0.0604
p=0.1566
p=0.6136
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DGCR8 | 91617_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.034
p=0.8024
p=0.1835
p=0.6382
SmallAirway2
LargeAirway2
Trachea
Macrophage2

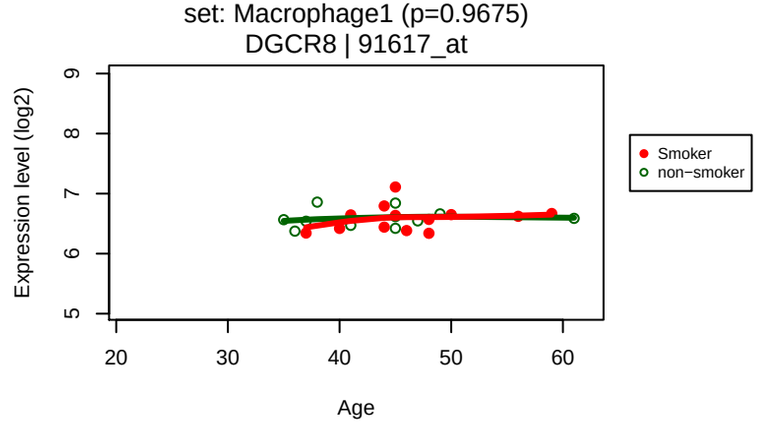

set: SmallAirway1 (p=1e−04)
DGCR8 | 91617_at
Expression level (log2)
Age
Smoker
non−smoker

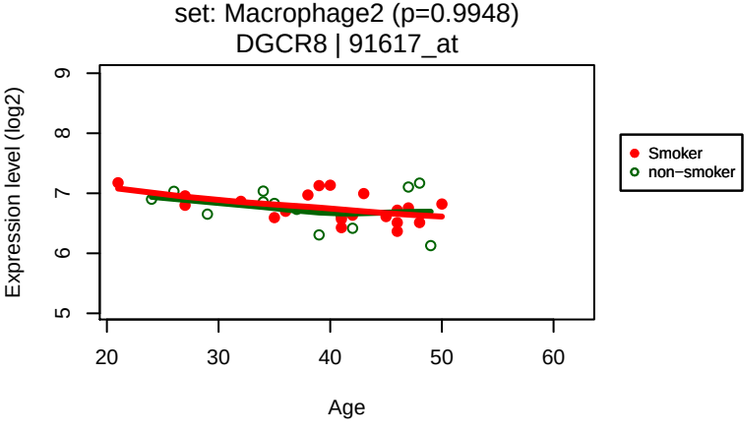

set: SmallAirway2 (p=0.0191)
DGCR8 | 91617_at
Expression level (log2)
Age
Smoker
non−smoker

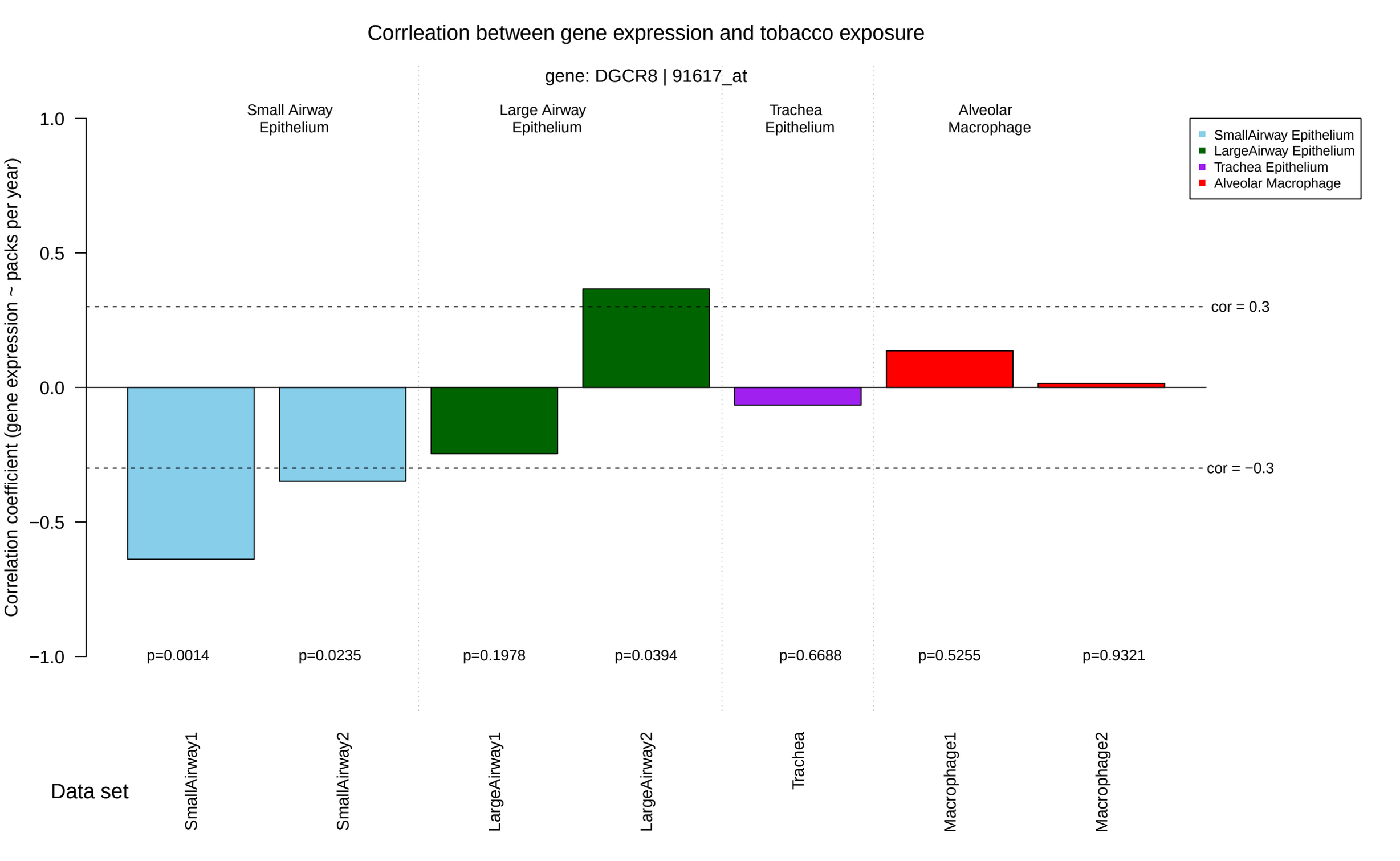

set: LargeAirway1 (p=0.3993)
DGCR8 | 91617_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.059)
DGCR8 | 91617_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3909)
DGCR8 | 91617_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9675)
DGCR8 | 91617_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9948)
DGCR8 | 91617_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DGCR8 | 91617_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0014
p=0.0235
p=0.1978
p=0.0394
p=0.6688
p=0.5255
p=0.9321
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2